

Project Abstract

EduPredict: Data-Driven Prediction of Student Academic Performance

Abstract:

In the evolving landscape of higher education, the ability to predict student success and intervene early is a significant challenge for educational institutions. This project, EduPredict, develops a machine learning framework to forecast academic performance and identify at-risk students using behavioral and academic data. Utilizing a synthetic dataset representing the 2026 SRM Institute of Science and Technology student cohort, the study incorporates various predictors...

The methodology involves comprehensive data preprocessing, feature engineering (including a weighted performance index), and the application of supervised learning models. Results demonstrate that ensemble methods like Random Forest provide superior predictive accuracy (~92%) compared to traditional logistic regression. The feature importance analysis highlights that early internal assessment scores and consistent attendance are the primary drivers of final academic grades. This research provides a scalable solution for educational analytics, enabling proactive student support systems that align with SDG Goal 4 (Quality Education). By automating the identification of underperforming students, institutions can implement targeted mentorship programs, ultimately improving retention rates and academic outcomes for the Data Science AIML-B 2026 cohort and beyond.